

(1)

DATE 04/01/2024

Physics

$$1 \quad \frac{pl}{A}, \frac{pl'}{A'}$$

$$\frac{l}{l'} = \frac{3}{1}, \quad l = 3l'$$

$$\frac{A}{A'} = \frac{2}{1}, \quad A = 2A'$$

$$\frac{pl}{2A'} = \frac{3:2}{2A'}$$

$$\frac{pl}{2A'}$$

$$\frac{pl}{A}$$

∴ option 3 - 3:2

$$2 \quad 4 \times 100 = 400 \text{ W}$$

$$400 \times 6 = \overset{2400}{2400} \text{ W per day}$$

$$\text{now, } 2400 \times 30 = 72000 \text{ W in 30 days or 72 KW}$$

$$\boxed{72000} \Rightarrow 72 \times 5 = 360$$

∴ option 1 = ₹360

3 1 - convergent lens

4 Myopia (Near sightedness)

We can use a concave lens to correct it
~~Normal~~

Hyperopia (far sightedness)

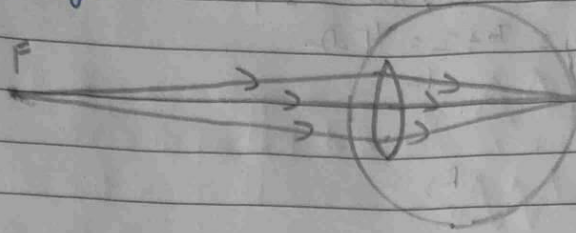
We use a convex lens to correct it

2

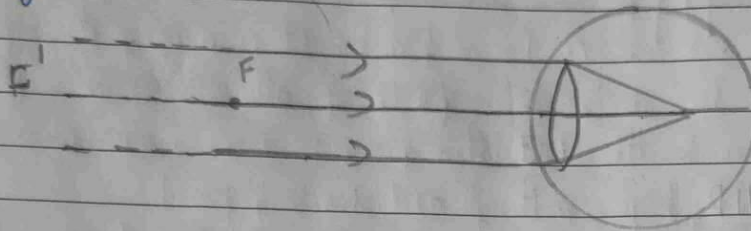
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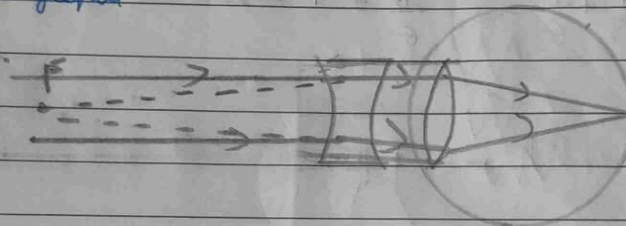
Far point of Myopic eye -



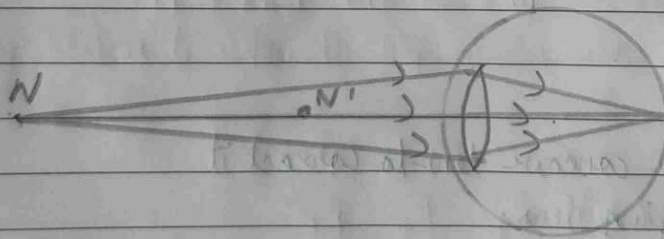
Myopic eye -



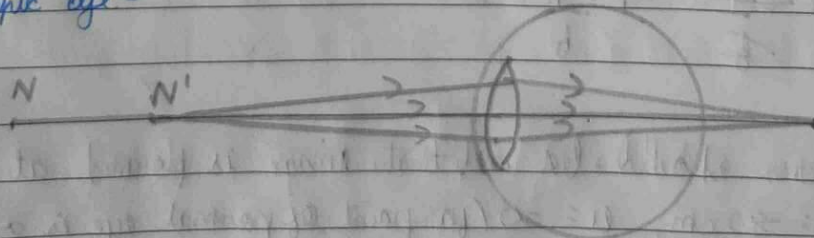
Correction for myopia -



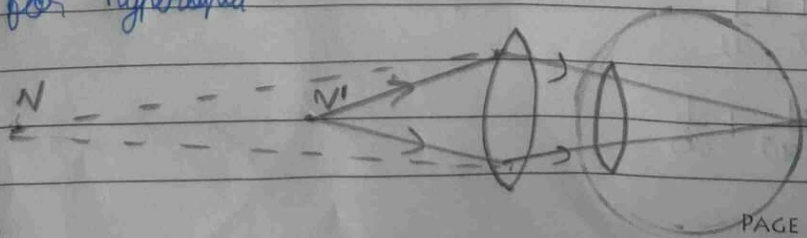
Near point of Hypermetropic eye -



Hypermetropic eye -



Correction for Hyperopia -



(3)

DATE

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5

i

Ammeter -

$$V = IR$$

$$I = \frac{V}{R}, \quad V = \frac{6}{20} + 2 + 2 = 6V$$

$$R = 3 + 8 + 12 = 25 \Omega$$

$$\frac{6}{25} = 0.24 A$$

$\therefore$ The reading will be 0.24 A in ammeter

Voltmeter -

$$V = IR$$

$$I = 0.24 A$$

$$R = 12 \Omega$$

$$0.24 \times 12 = \frac{24 \times 12}{100} = 2.88 V$$

$\therefore$ The reading will be ~~2.88 V~~ 2.88 V in voltmeter

ii

6

$$4 \Omega -$$

$$\text{Let } R_1 = 2 \Omega, R_2 = 3 \Omega, R_3 = 6 \Omega$$

When $R_2 \parallel R_3 = R'$, R' in series to R_1

$$\frac{1}{R'} = \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{R'} = \frac{1}{3} + \frac{1}{6}$$

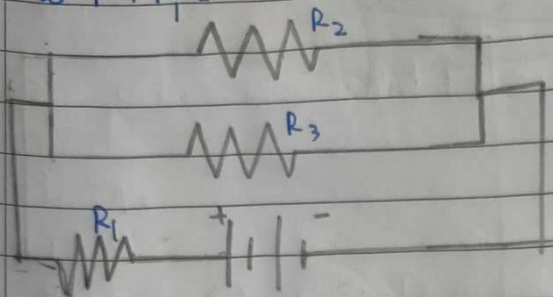
$$= \frac{2+1}{6} = \frac{3}{6}, \quad R' = 2 \Omega$$

classmate

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Now R' in series to R_1

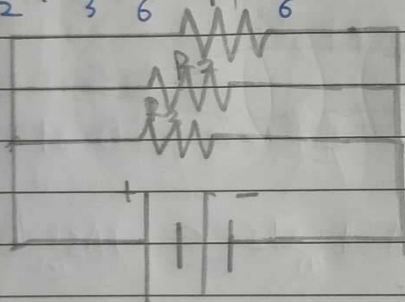
$\therefore R' + R_1 = 2 + 2 = 4 \Omega$



2 1Ω

$R_1 \parallel R_2 \parallel R_3$

$= \frac{1}{\frac{1}{2} + \frac{1}{3} + \frac{1}{6}} = \frac{6}{6} = 1 \Omega$



6

- i We will use a concave lens to correct it
Nature = diverging lens

~~Power~~ $= \frac{1}{f}$ $P = \frac{1}{f}$

The lens should be such that image is formed at far point
so $v = -80 \text{ cm}$, $u = \infty$ (far point of normal eye is ∞)

$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

$= \frac{1}{f} = \frac{1}{-80} + \frac{1}{\infty}$

classmate

$$\frac{1}{f} = \frac{-1}{80} + 0$$

~~$$P = \frac{1}{f}$$~~
$$f = -80 \text{ cm or } -0.8 \text{ m}$$

~~$$P = \frac{-1}{80}$$~~

$$P = \frac{1}{f}$$

=

$$\frac{1}{-0.8}$$

=

$$= \frac{-10^5}{-84} = -1.25$$

~~$$P = -8$$~~

∴ Power of lens required = -1.25

ii The ability of the eye lens to adjust its focal length using ciliary muscles is power of accommodation

iii Iris gives color to eye
Regulates size of pupil

Q

A